Tharindu Ekanayake

Computer Vision Engineer

Duisburg, Germany | tharindu326@gmail.com
tharindu326.github.io | github.com/tharindu326 | linkedin.com/in/tharindu-ekanayake
Work authorization: EU Blue Card holder

PROFESSIONAL SUMMARY

Computer Vision Engineer with 5+ years building production perception systems spanning model development, multi-camera vision, GPU optimization and deployment. Experienced in detection, segmentation, tracking, re-identification and pose estimation across real-time video, edge and GPU infrastructure, with additional depth in 3D pose, LiDAR perception and human sensing. Currently developing production GPU video inference services at goAVA using CUDA, Triton and batched pose estimation. Previously built the perception stack for an autonomous checkout system used in live retail environments. Published computer vision research at IEEE ICASSP and SCIA, including first-author work in 3D pose canonicalization and camera-based human sensing.

TECHNICAL SKILLS

Detection & Segmentation: YOLO11, RT-DETR, Mask R-CNN, SAM2, Sapiens, YOLO segmentation

Tracking & Pose: ByteTrack, OC-SORT, StrongSORT, DeepSORT, re-ID, ViTPose, YOLO-Pose, OpenPose, MotionBERT

3D & Multi-Camera: PV-RCNN, OpenPCDet, Open3D, point clouds, calibration, triangulation, identity association

GPU & Edge Deployment: CUDA, CuPy, TensorRT, FP16, NVENC, ONNX, NVIDIA Triton, NVIDIA Jetson, Coral TPU

ML, Vision & Signal Processing: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV, PySceneDetect, scikit-image, SciPy, rPPG, FFT, STFT, Butterworth and FIR filtering

MLOps & Serving: Docker, Kubernetes (GKE), FastAPI, Hugging Face TEI, GitHub Actions, Weights & Biases

LLM & RAG: hybrid retrieval (multilingual-E5, SPLADE, RRF), cross-encoder reranking, Qdrant, LLM-as-judge

Programming: Python, SQL, Bash

WORK EXPERIENCE

Machine Learning Engineer, goAVA GmbH

Essen, Germany | 01/2026 – Present

- Developed and deployed production GPU computer vision services for video files and RTSP streams using batched ViTPose and YOLO11-Pose inference served through NVIDIA Triton with CUDA and FP16.
- Built CUDA-accelerated video frame interpolation using custom CuPy kernels, FP16 processing and NVIDIA NVENC hardware encoding, with adaptive processing based on input resolution.
- Developed video and image processing pipelines using OpenCV, PySceneDetect, scikit-image, SciPy, Pillow and VLM-based image understanding, integrated into the production ingestion pipeline.
- Built a production multimodal ML platform processing documents, audio, video, images and tabular data, moving GPU model inference to dedicated Hugging Face TEI services (~0.4 to 2.6 req/s, ~6.5x, on a single NVIDIA L4) and improving retrieval recall@5 from 0.78 to 0.90.
- Implemented production deployment infrastructure using Docker, Kubernetes, nginx and GitHub Actions, with automated testing, CI-enforced regression checks and security scanning.

Project Researcher, CMVS, University of Oulu

Oulu, Finland | 10/2024 – 01/2026

- Designed 3DPCNet, a hybrid GCN-Transformer for viewpoint-invariant 3D pose canonicalization with adaptive graph learning, cross-attention fusion and 6D rotation representation, reducing MPJPE by ~35% from 74.1 mm to 48.4 mm in cross-environment evaluation; first-author IEEE ICASSP 2026 paper.
- Adapted OpenPCDet and PV-RCNN for sparse indoor LiDAR person detection, including keypoint-derived 3D boxes, voxelization, coordinate-frame alignment and KITTI-format conversion, enabling a privacy-preserving HAR pipeline that reached 94.4% accuracy on 13 rehabilitation activities and 90.3% across all 27 MM-Fi activities under cross-scene, subject-independent evaluation.

- Built camera-based human perception pipelines for VR training from monocular RGB video, combining person detection, tracking, pose estimation, segmentation, 3D kinematics and rPPG signal extraction.
- Developed semi-automatic annotation pipelines using SAM2, YOLO segmentation, ViTPose, Sapiens and multiple trackers, with training and evaluation on CSC Finland's Puhti and Mahti HPC clusters.

Machine Learning Engineer, ZOOMi Technologies Inc. *Colombo, Sri Lanka | 10/2020 – 09/2023*

- Built the perception stack for PixelMagic, an autonomous checkout system used in live retail environments, covering object detection, segmentation, multi-object tracking, re-identification and pose estimation across synchronized cameras.
- Developed multi-camera tracking that maintained stable shopper identities through occlusion and viewpoint changes, using triangulation to resolve person positions across overlapping views.
- Optimized computer vision models for NVIDIA Jetson using TensorRT, increasing inference performance from ~12–15 FPS to ~28–30 FPS while operating within edge memory and compute constraints.
- Engineered real-time multiprocess video pipelines that kept live camera streams within frame budgets and handled failure cases encountered in real retail environments.
- Contributed to LayerNext, a computer vision data and training platform covering video ingestion, annotation and dataset versioning, and delivered computer vision systems for medical imaging and autonomous agricultural navigation.

PROJECTS

PhysioTrack, Contactless Human-Sensing Library *Open source | 05/2025 – Present*

- Maintain a pip-installable GPL-3.0 Python library (github.com/tharindu326/physiotrack) unifying detection, tracking, 2D/3D pose, segmentation, monocular depth, face analysis and rPPG behind a single prediction API.
- Integrated 51 pretrained model variants, including YOLO11/12, RT-DETR, ViTPose, Sapiens, MotionBERT, 3DPCNet and multiple tracking algorithms, with a model registry and on-demand Hugging Face weight resolution cached for Docker deployments; built MkDocs documentation, pytest coverage and semantic-release CI via GitHub Actions.

EDUCATION

M.Sc. Computer Science and Engineering, Artificial Intelligence *Finland | 09/2023 – 05/2025*

University of Oulu

Thesis: Unsupervised multimodal biosignal analysis of student cognitive and physiological responses during VR-based training sessions | *Grade:* 5/5 (highest)

B.Sc. Electrical and Electronic Engineering *Sri Lanka | 2016 – 04/2020*

University of Peradeniya

PUBLICATIONS

- **T. Ekanayake**, C. Á. Casado, M. B. López. 3DPCNet: Pose Canonicalization for Robust Viewpoint-Invariant 3D Kinematic Analysis from Monocular RGB Cameras. *IEEE ICASSP*, 2026. First author. DOI: 10.1109/ICASSP55912.2026.11464754
- S. Sharifipour, C. Á. Casado, L. Nguyen, **T. Ekanayake**, M. L. Cañellas, et al. LiDAR-Based Human Activity Recognition through Laplacian Spectral Analysis. *IEEE ICASSP*, 2026. Co-author. DOI: 10.1109/ICASSP55912.2026.11462769
- **T. Ekanayake**, C. Á. Casado, N. Nguyen, M. Sobocinski, et al. Evaluating the Accuracy and Reliability of Camera-Based Physiological and Motion Signal Extraction Techniques in Virtual Reality Training Environments. *SCIA*, *Springer LNCS*, 2025. First author. DOI: 10.1007/978-3-031-95911-0_31